MODBUS RTU REFERENCE

STANDARD COMMANDS AND PROTOCOL SPECIFICATIONS

Modbus RTU (Remote Terminal Unit) is a serial communications protocol widely used in industrial automation systems. It operates on a master/slave (client/server) architecture where a single master initiates all transactions.

1. Standard Data Model

The Modbus protocol defines four primary data tables for reading and writing device values. Access rights dictate whether a master can read or write these elements.

Primary Table	Object Type	Access Type	Provided By
Discrete Inputs	Single bit (0 or 1)	Read-Only	I/O system (Digital Inputs)
Coils	Single bit (0 or 1)	Read-Write	Application Program (Digital Outputs)
Input Registers	16-bit word	Read-Only	I/O system (Analog Inputs)
Holding Registers	16-bit word	Read-Write	Application Program (Analog Outputs/Settings)

2. RTU Message Frame Format

Modbus RTU messages are transmitted as a continuous stream of bytes. Frames are separated by a silent interval of at least 3.5 character times (t3.5).

Slave Address	Function Code	Data payload	CRC Checksum
1 Byte (1-247)	1 Byte (Command)	0 to 252 Bytes (Addresses, Values)	2 Bytes (Cyclic Redundancy Check)

3. Read Function Codes

Code (Dec)	Code (Hex)	Function Name	Description
01	`0x01`	Read Coils	Reads the status of one or more contiguous coils (digital outputs) from a slave device.
02	`0x02`	Read Discrete Inputs	Reads the status of contiguous discrete inputs (digital inputs) from a slave.
03	`0x03`	Read Holding Registers	Reads the contents of contiguous block of holding registers (configuration/analog out).
04	`0x04`	Read Input Registers	Reads the contents of contiguous block of input registers (analog inputs).

4. Write & Diagnostic Function Codes

Code (Dec)	Code (Hex)	Function Name	Description
05	0x05	Write Single Coil	Writes a single output to ON (0xFF00) or OFF (0x0000).
06	0x06	Write Single Register	Writes a single 16-bit value to a holding register.
08	0x08	Diagnostics	Provides a series of tests for checking the communication system (e.g., Echo test).
15	0x0F	Write Multiple Coils	Forces a sequence of coils to either ON or OFF state.
16	0x10	Write Multiple Registers	Writes a block of contiguous holding registers to specific values.

5. Standard Exception Codes

If a slave receives a valid request but cannot perform the requested action, it returns an exception response. The function code in the response is set to the original function code + 0x80 (e.g., 0x03 becomes 0x83).

Code	Exception Name	Description / Meaning
01	Illegal Function	The function code received in the query is not an allowable action for the slave device.
02	Illegal Data Address	The data address received in the query is not an allowable address for the slave. It falls outside the configured memory map.
03	Illegal Data Value	A value contained in the query data field is not an allowable value for the slave (e.g., requesting 200 registers when the limit is 125).
04	Slave Device Failure	An unrecoverable error occurred while the slave was attempting to perform the requested action.

Request Example: Read Holding Register (0x03)

Master Request: 01 03 00 6B 00 03 76 87

- 01 : Slave Address 1
- 03 : Function Code (Read Holding Reg)
- 00 6B : Starting Address (107)
- 00 03 : Quantity of Registers (3)
- 76 87 : CRC Checksum